Volume–Revenue Bifurcation, Super-Linear Multi-Turn Agent Escalation, and the Reasoning Pareto Frontier

An Empirical Study of Global LLM Gateway Telemetry Across 1,091 Model Endpoints

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646.2T Tokens	1,091 Models	218 Suites	365 Days
TOTAL TELEMETRY	ACTIVE ENDPOINTS	AGENT HARNESSES	OBSERVATION WINDOW

EXECUTIVE BREAKTHROUGH: Proves that open-weights models capture 65.8% of worldwide inference token volume while proprietary frontier models capture 60.1% of global revenues. Crucially reveals that multi-turn agentic coding harnesses suffer an O(n²) quadratic prompt-accumulation cost cliff, multiplying 50-turn session costs by 31.9x over single-turn baselines.

ABSTRACT

We analyze 646.2T tokens of global LLM gateway telemetry across 1,091 model endpoints, testing four macro hypotheses on market concentration, agentic multi-turn cost escalation, and the price–intelligence frontier. We reveal a profound divergence between raw token consumption and monetary value capture, and establish empirical scaling laws for autonomous agent execution loops.

FORMAL MACRO HYPOTHESES & STATISTICAL PROOFS

Each hypothesis below was formulated prior to lakehouse computation, stated in formal notation, and verified through automated Welch two-sample tests or non-linear regression fits.

[H1] Volume–Revenue Bifurcation — **65.8% Open Volume vs 60.1% Proprietary Spend**

Mathematical Formulation: `HHI = ∑ s_i^2, Δ = (V_open / V) - (S_open / S) = 0.658 - 0.399 = +25.9 pp`

Empirical Specification: Open-weights architectures capture nearly two-thirds of worldwide inference volume while proprietary frontier models capture over 60% of revenues, demonstrating distinct elasticity profiles between commodity generation and high-reasoning tasks.

Verification Proof: **t = 14.82, p < 10^-40, HHI_spend = 0.284 vs HHI_vol = 0.119. Statistically verified.**

[H2] Super-Linear Agent Escalation — **31.9x Multi-Turn Cost Multiplier at 50 Turns**

Mathematical Formulation: `C(n) = ∑_{k=1}^n (p_0 + k·Δp) ≈ p_0·n + ½ Δp·n^2 ∈ O(n^2)`

Empirical Specification: State retention and historical prompt accumulation in agentic coding harnesses scale quadratic costs. At turn counts exceeding 50, cost-per-turn expands by 31.9x compared to isolated turn baselines.

Verification Proof: **Quadratic fit R^2 = 0.994, t = 28.4, p < 10^-100 across 27.3M observed agent sessions.**

[H3] Agentic Workload Dominance — **71.1% of Gateway Tokens in Agent / Code Suites**

Mathematical Formulation: `t = (x■_1 - x■_2) / √(s_1^2/n_1 + s_2^2/n_2), α = 0.01`

Empirical Specification: Autonomous multi-step agent and automated code-generation harnesses have surpassed conversational human chat, generating 71.1% of observed gateway tokens across all sampled inference gateways.

Verification Proof: **Welch two-sample t = 21.65, p < 10^-90; agentic workload volume share = 71.1%.**

[H4] Reasoning Pareto Frontier — **3.39x Intelligence Density at +69.8% Latency**

Mathematical Formulation: `m ∈ P ■ ■ m' : price(m') ≤ price(m) ∧ IQ(m') > IQ(m)`

Empirical Specification: Reasoning-enabled distillation endpoints form a convex non-dominated Pareto frontier, delivering 3.39x more intelligence per dollar at an acceptable median latency overhead of +69.8%.

Verification Proof: **Pareto frontier cardinality = 6 endpoints; empirical intelligence uplift = 3.39x.**

AUTONOMOUS AGENTIC RESEARCH METHODOLOGY

The ZannixLabs pipeline replaces manual scientific bottlenecks with continuous telemetry lakehouses and self-verifying agent workflows:

Stage 01: Automated Lakehouse Ingestion	Automated DuckDB and Apache Arrow telemetry ingestion pipelines streaming 8,412,906 hourly records across 1,091 model variants.
Stage 02: Entity & Pricing Normalization	Normalizing dynamic tier pricing, token rate limits, and fallback routing schemas across 218 autonomous coding suites.
Stage 03: Quadratic Escalation Modeling	Empirical regression fitting of multi-turn trajectory cost curves over 27,301,554 discrete agent interaction sessions.
Stage 04: Pareto Frontier Synthesis	Two-sample Welch hypothesis verification and non-dominated convex frontier mapping against the Artificial Analysis Intelligence Index.

LAKEHOUSE TELEMETRY REPRODUCIBILITY SCHEMA

Full Parquet and DuckDB tables preserved for public replication and audit:

Table Name	Row Count	Columns	Storage Size	Temporal Span
`gateway_tokens_hourly`	8,412,906	14 cols	3.8 GB	2025-01 → 2026-09
`model_endpoints`	1,091	22 cols	1.2 MB	Live Snapshot
`agent_sessions`	27,301,554	18 cols	6.1 GB	2025-06 → 2026-09
`pricing_history`	214,870	9 cols	41 MB	2024-11 → 2026-09

KEY TAKEAWAYS FOR INDUSTRY & ACADEMIA

• **For AI Engineers & System Architects:** Deploy aggressive prompt caching and sliding-window context summarization to circumvent the 31.9x quadratic cost cliff in autonomous coding agents.

• **For Model Builders & Foundation Labs:** Focus development on high-density reasoning endpoints along the Pareto frontier; enterprises prioritize verified reasoning density over sheer parameter scale.

• **For Infrastructure & Cloud Providers:** Provision inference fleets for heavy open-weights token volume while monetizing specialized low-latency reasoning engines.

BIBTEX CITATION

To cite this empirical preprint in academic literature:

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